Superposition and Probability Amplitudes

Why Superposition Matters

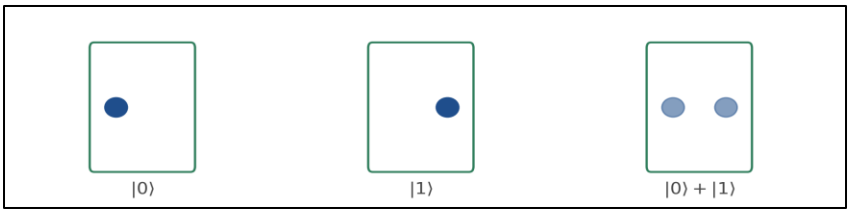
Imagine searching for a friend in a city. A classical computer follows one road at a time: Road A, then Road B, then Road C, until it finds the answer. A quantum computer works differently. Instead of committing to one possibility immediately, it can represent multiple possibilities within a quantum state. This ability comes from a fundamental principle called superposition. Without superposition there would be no quantum computing, no quantum algorithms, and no quantum advantage.

The Classical World

A classical bit can only be 0 or 1, nothing in between, like a light switch that is ON or OFF, or a door that is open or closed. Classical information is always definite.

The Quantum World

A qubit behaves differently. Before measurement it can be |0>, |1>, or a combination of both. This combination is called superposition.



1. What Is Superposition

Superposition means a qubit can exist in a quantum state that contains contributions from multiple basis states simultaneously. For one qubit this is written as |psi> = alpha|0> + beta|1>, which is called a superposition state.

Analogy: Music

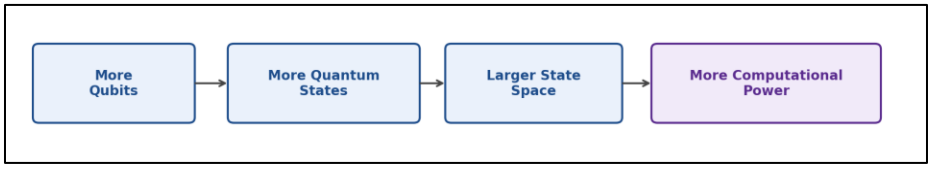
Pressing one piano key produces one note. Pressing two keys together produces a sound containing information from both notes. Similarly, a superposition state contains information from multiple basis states. This analogy is closer to reality than saying a qubit is 0 and 1 at the same time.

The Most Famous Superposition State

Consider |psi> = (|0> + |1>) / square root of 2. This is called equal superposition, where both possibilities contribute equally to the quantum state, giving a 50 percent chance of measuring either outcome.

Why Superposition Is Powerful

With 1 qubit there are 2 possible states, |0> and |1>. With 2 qubits there are 4 possible basis states: |00>, |01>, |10>, and |11>. A quantum system can exist in a superposition of all these basis states at once.

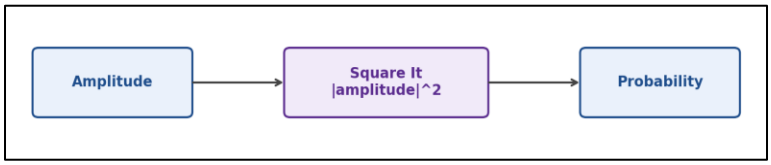


2. Probability Amplitudes

Most beginners think probability and amplitude are the same, but they are not. Probability tells us how likely an outcome is. Amplitude is the mathematical value inside a quantum state that is used to calculate that probability.

The Born Rule

The rule for converting amplitudes into probabilities is Probability = |Amplitude|^2, known as the Born Rule, one of the most important rules in quantum mechanics.



Two Examples

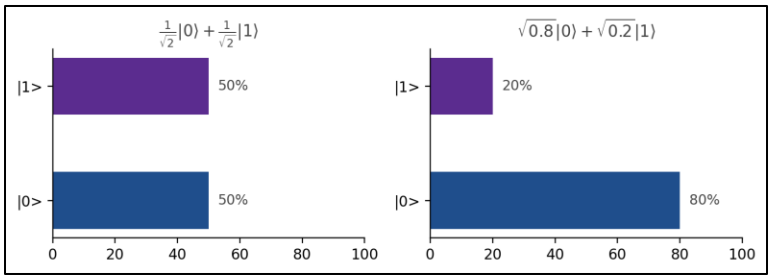


Figure 1: Equal superposition gives a 50/50 split; unequal amplitudes give unequal probabilities.

Every valid quantum state must satisfy |alpha|^2 + |beta|^2 = 1, because measurement must produce either |0> or |1>, and the total probability must always equal 100 percent.

3. Phase and Why Amplitudes Matter

Probability amplitudes can be positive, negative, or complex, which introduces quantum phase. The states (|0>+|1>)/root2 and (|0>-|1>)/root2 both give a 50/50 split after measurement, yet they behave differently inside quantum algorithms. This difference comes from phase, and phase is what enables quantum interference.

Quantum algorithms do not directly manipulate probabilities, they manipulate probability amplitudes, amplifying the amplitudes of correct answers and suppressing incorrect ones. When measurement finally occurs, the correct answer becomes more likely. This idea powers many quantum algorithms, similar to a campaign that shifts votes from an even 50/50 split toward one clear winner.

Quick Comparison

Concept	Meaning
Superposition	Combination of multiple basis states
Probability amplitude	Mathematical value inside a quantum state
Probability	Chance of measuring an outcome
Born Rule	Probability = |Amplitude|^2

Concept	Meaning
Measurement	Converts a quantum state into a classical outcome
Phase	Hidden property that affects interference

Summary: What to Remember

Superposition lets a qubit hold contributions from multiple basis states at once

Equal superposition, (|0>+|1>)/root2, gives a 50 percent chance of either outcome

Probability amplitudes are not the same as probabilities

The Born Rule converts amplitudes into probabilities: Probability = |Amplitude|^2

Every valid state must satisfy |alpha|^2 + |beta|^2 = 1

Phase can differ between states that have the same measurement probabilities

Quantum algorithms amplify correct answers and suppress incorrect ones before measurement